EEMT 5510
Engineering Economics and Cost Management

Chapter 8
Dealing with Uncertainty and Multi-attributes

Huifu Xu
Dept. of Systems Engineering & Engineering Management
The Chinese University of Hong Kong

Uncertainty

- Future outcome of a project may not be deterministic
  - Revenue, cost, useful life, etc.
- Renting versus purchasing a home
  - A common situation faced by young families is whether to rent or buy a home. Many of the decision factors are speculative, such as the future resale value of the house and how long the family will be in a particular area.

Uncertainty

- Future outcome of a project may not be deterministic
  - Revenue, cost, useful life, etc.
- Two approaches
  - Probability may not be known
    - Breakeven analysis
    - Sensitivity analysis
    - Scenario analysis
  - Probability for each possible future outcome may be known
    - The average performance

Beyond PW analysis

- Suppose the PW analysis says we should accept project X. Should we really accept X?
- Two possibilities:
  - (1) PW is really positive.
  - (2) PW is positive because of inaccurate estimation.
- Rules to remember:
  - (1) positive PW projects are rare, especially in a highly competitive environment.
  - (2) there often exist some uncertainties.

Breakeven and Sensitivity Analysis

- The motivation for breakeven and sensitivity analysis is to establish the bounds of error in our estimates such that another alternative being considered may become a better choice than the one we recommended under assumed certainty.
- Breakeven and Sensitivity Analysis are nonprobabilistic techniques for tackling risk and uncertainty.

Breakeven Analysis

- Breakeven analysis determines the value of a critical factor at which economic trade-offs are balanced.
- When the selection between two engineering project alternatives (or outcomes) is heavily dependent on a single factor, we can solve for the value of that factor at which the conclusion is a standoff.
- That value is known as the *breakeven point*, that is, the value at which we are indifferent between the two alternatives.

Example 8.1: Breakeven analysis

- A company is considering to launch a new product.
  - Initial cost $200,000.
  - Five year study period. No salvage value.
  - MARR=10%.
  - The project estimates are

Unit sales	6,000
Price per unit	80
Variable cost per unit	60
Fixed cost	60,000

- PW(Q) =-$200,000+[(80-60)Q-60,000](P/A,10%,5)
  - For Q = 6000, PW=27,477.

Example 8.1: Breakeven analysis

- Breakeven analysis of Q:
  - find out the level of Q such that $PW(Q^*)=0$.
  - $Q^* = [200{,}000(A/P, 10\%, 5) + 60{,}000]/20 = 5638$
- If the actual $Q \geq Q^*$, then the project is profitable, otherwise it is not.

Example 8.2: Breakeven analysis for 2 options

- Two options
  - Variable factor (X): the number of operating hours per year
- Alternative A
  - Capital investment $12,500 with 10 years life
  - Annual maintenance cost $687
  - Annual operating cost $5.04X
- Alternative B
  - Capital investment $16,000 with 10 years life
  - Annual maintenance cost $490
  - Annual operating cost $4.05*X*
- Find the breakeven point at MARR=15%

Example 8.2: Breakeven analysis for 2 options

- Annual cost for Alternative A

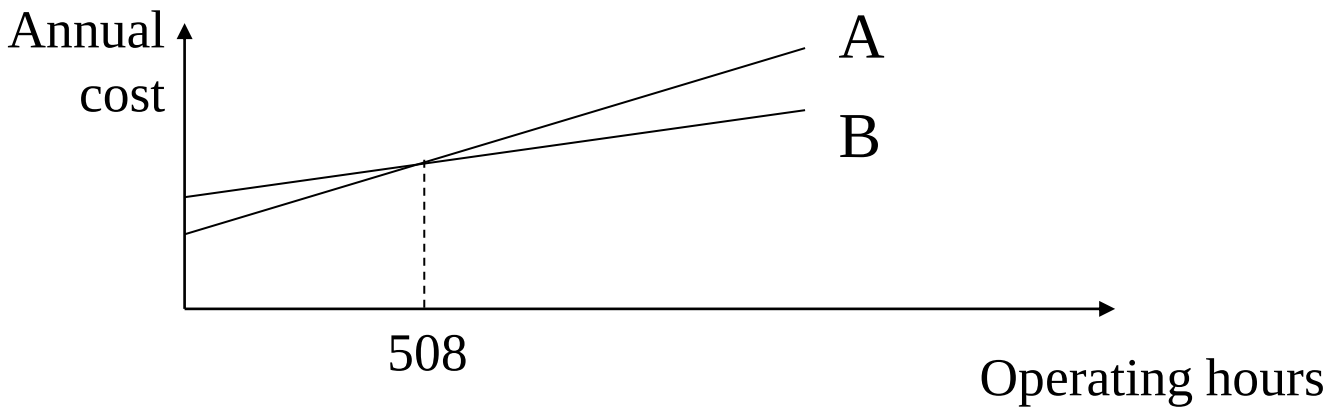
$$AW_A = 12{,}500(A/P,15\%,10) + 687 + 5.04X$$

- Annual cost for Alternative B

$$AW_B = 16{,}000(A/P,15\%,10) + 490 + 4.05X$$

- Setting $AW_A = AW_B$, we get the breakeven point

$$X = 508 \text{ hours}$$



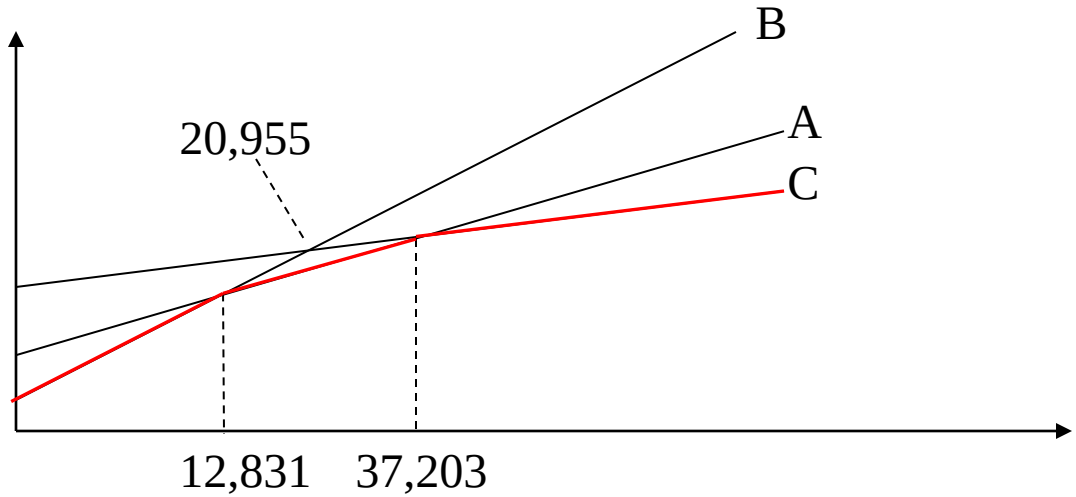
Example 8.3: Breakeven analysis for 3 options

- An equipment to be installed on trucks to save fuel consumption
  - Annual fuel cost depends on the mileage (X) for which a truck is to be used
- Repeatability assumption
- MARR=10%

	Option A	Option B	Option C
Initial Investment	1000	400	1200
Annual maintenance cost	10	5	5
Annual fuel cost	$0.184X$	$0.192X$	$0.18X$
Useful life (years)	10	10	5

Example 8.3: Breakeven analysis for 3 options

- $AW_A = 1{,}000(A/P,10\%,10)+10+0.184X = 172.75+0.184X$
- $AW_B = 400(A/P,10\%,10)+5+0.192X = 70.1+0.192X$
- $AW_C = 1{,}200(A/P,10\%,5)+5+0.180X = 321.56+0.180X$
- Breakeven points
  - For (A,B): 12,831; for (A,C): 37,203; for (B,C): 20,955



Sensitivity analysis

- Goal:
  - To identify variables to which the economic profitability of a project is sensitive to.
- Idea:
  - Pick one variable. Freeze all other variables and see how sensitive our estimates of PW is to changes in the chosen variable. Repeat this procedure for all other variables.
- Definition of sensitivity for a variable:
  - the percentage change in that variable that will reverse the decision.
- For those sensitive variables, we need to spend more efforts in estimating their value.

Example 8.4

- Consider a proposal to enhance the vision system used by a postal service to sort mail. The new system is estimated to cost $1.1 million and will incur an additional $200,000 per year in maintenance costs. The system will produce annual savings of $500,000 each year (primarily by decreasing the percentage of misdirected mail and reducing the amount of mail that must be sorted manually). The MARR is 10% per year, and the study period is 5 years at which time the system will be technologically obsolete (worthless).

Example 8.4

- PW = -1,100,000+(500,000-200,000)(P/A,10%, 5) = $37, 236
- Let x be the break-even percentage change in annual savings (i.e. PW=0).
  - -1,100,000 + [500,000(1 + x)-200,000](P/A,10%,5) = 0
  - x = -2.0%
- If the estimated annual savings is lower by more than 2% of its most likely value, the decision about the project is reversed (from accept to reject).

Example 8.4

- The sensitivity for other variables can be calculated similarly.
- Below is the summary.

Capital investment	3.4%
Annual savings	2.0%
Annual expense	4.9%
MARR	13.2%

Example 8.5

- A project with the following issues to be considered
  - Capital investment: $11,500
  - Annual profit: $3,000
  - Salvage value: $1,000
  - Useful life: 6 years
- PW at MARR=10%

  $-11{,}500+3{,}000(P/A,10\%,6)+1{,}000(P/F,10\%,6)=\$2{,}130$
- What is the change of PW if
  - The capital investment changes, or
  - The annual profit changes, or
  - The salvage value changes

Example 8.5

- If the capital investment changes by *p*%

$$PW_p = -11500(1+p\%)+3000(P/A,10\%,6)+1000(P/F,10\%,6)$$
$$=2130-11500(p\%)$$

- If the annual profit changes by *a*%

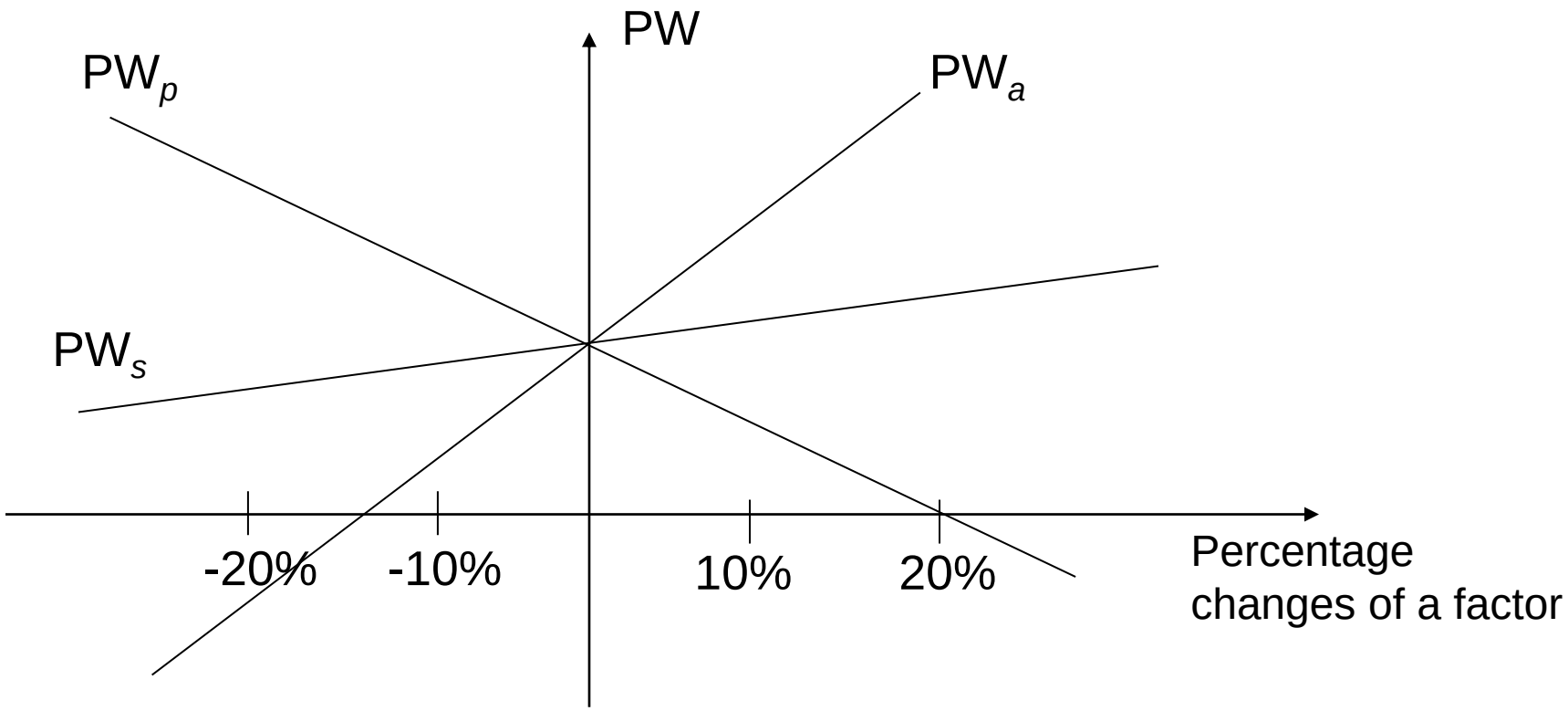
$$PW_a = -11500+3000(1+a\%)(P/A,10\%,6)+1000(P/F,10\%,6)$$
$$= 2130+13065(a\%)$$

- If the salvage value changes by *s*%

$$PW_s = -11500+3000(P/A,10\%,6)+1000(1+s\%)(P/F,10\%,6)$$
$$= 2130+564(s\%)$$

Example 8.5

- Sensitivity graph



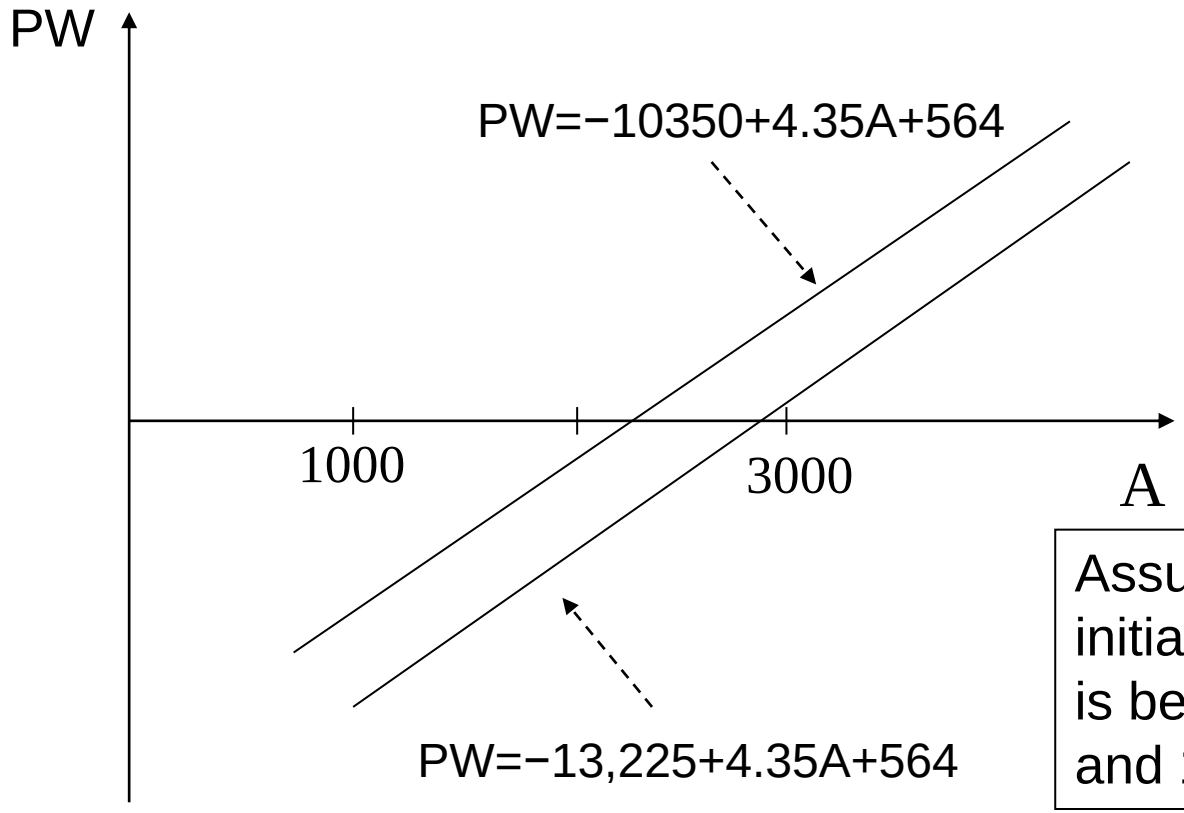
Question: the PW is more sensitive to which factor?

Combination of factors

- Handling the two most sensitive factors
  - Less sensitive factors are assumed to be fixed
- Regarding the given example
  - PW is more sensitive to initial investment (*I*) and annual profit (*A*)
  - Salvage value is fixed at $1000
  - PW becomes a function of two variables *I* and *A*, thus can be plotted on a graph

$$PW=I+A(P/A,10\%,6)+1000(P/F,10\%,6)=I+4.3553A+564$$

Combination of factors



Assuming the initial investment is between 10350 and 13225

PW=I+A(P/A,10%,6)+1000(P/F,10%,6)=*I*+4.3553*A*+564

Three or more uncertain factors

- Scenario analysis
  - For each factor, we identify 3 scenarios, the best, the worst, and the most-likely
  - The overall performance is evaluated for all possible combinations of scenarios

Example 8.6

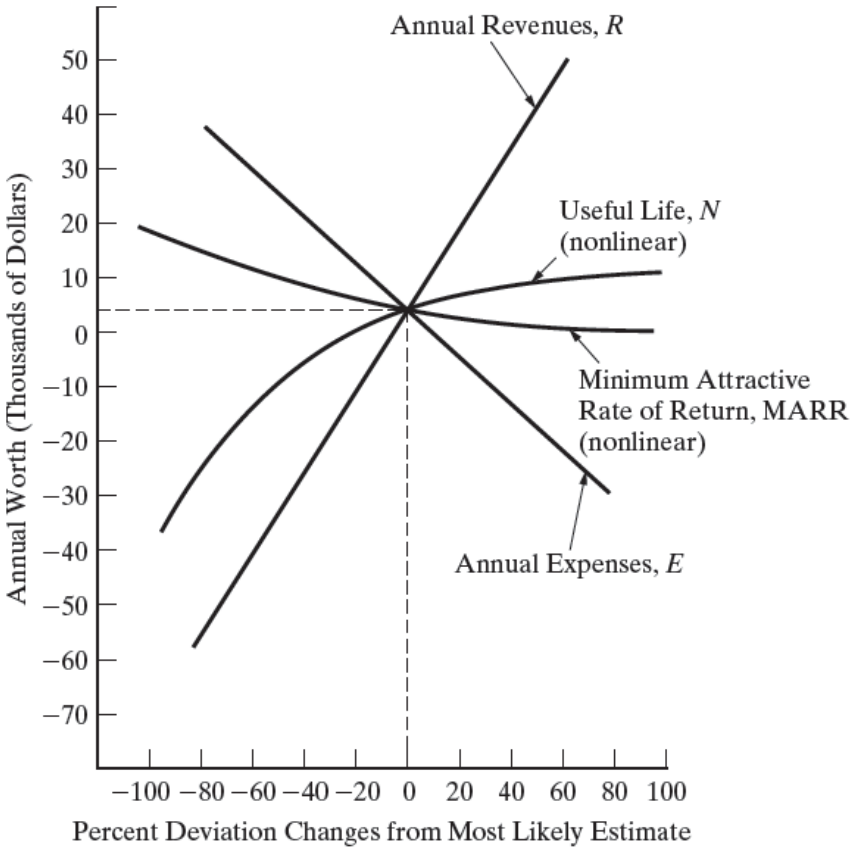
- A project has three uncertain factors
  - Useful life, annual revenue, and annual expense
  - Each factor is associated with 3 scenarios

Scenarios	Best	Most-likely	Worst
Investment	$150,000	$150,000	$150,000
Useful life	18	10	8
Annual Revenue	$110,000	$70,000	$50,000
Annual expense	$20,000	$43,000	$57,000
AW(8%)	$73,995	$4,650	−$33,100

Example 8.6

- Sensitivity analysis

Scenarios	Most-likely
Investment	$150,000
Useful life	10
Annual Revenue	$70,000
Annual expense	$43,000
AW(8%)	$4,650



Example 8.6

- For the overall performance, there are 27 scenarios
- The AW for each scenario can be evaluated

Annual Revenue	Annual Expense								
	Best Scenario			Most-likely			Worst Scenario		
	Useful Life			Useful Life			Useful Life		
	Best	M-L	Worst	Best	M-L	Worst	Best	M-L	Worst
Best	73995	67650	63900	50995	44650	40900	36995	30650	26900
M-L	34000	27650	23900	10995	4650	900	-3005	-9350	-13100
Worst	14000	7650	3900	-9005	-15350	-19100	-23005	-29350	-33100

Summary

- Breakeven analysis finds out the point where you break even. It allows you to be more confident about the result if you can compare the actual value and the breakeven value.
- Sensitivity analysis finds out the sensitive variables. More efforts need to be spent on estimating them.
- Scenario analysis shows the distribution of profitability over different scenarios.
- For a given problem, these methods can be used together to yield more convincing results.

Evaluating Projects by Probability Models

Probability models

- Randomness in a project
  - Cash flow for any time period may be a random variable
  - The useful life of a project may be a random variable
  - The interest rate may be a random variable
  - Assumptions: with known distributions
- Our concerns
  - The mean of PW, AW or FW
  - The variance of PW, AW or FW
  - Probability of (any of them < 0)
- Decision criteria
  - Mainly on the expected value, with additional considerations of variance and probability of some events

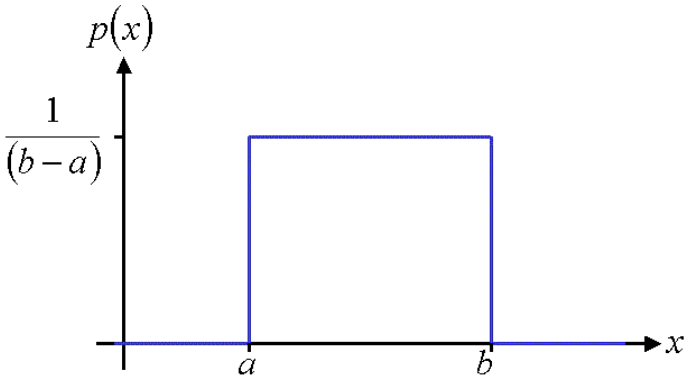
Random variables

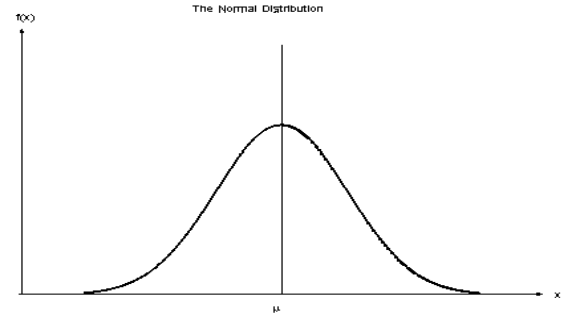
- Discrete random variables
  - Possible values are discrete $\{x_1, x_2, \ldots\}$
  - Described by a probability mass function $P(x_i)$, and cumulative distribution function
  - $P(x_i)$ is the probability of $X=x_i$
- Example: $P(X=1)=0.3$, $P(X=2)=0.3$, $P(X=3)=0.4$

x_i	1	2	3
$P(x_i)$	0.3	0.3	0.4

Random variables

- Continuous random variable
  - Possible values are in an interval [a, b]
  - Described by a probability density function $f(x)$, and cumulative distribution function
- Examples
  - Uniform distribution $Y \sim U[a, b]$
  - Normal distribution $Z \sim N(\mu, \sigma^2)$





Probability, Expectation and Variance

- Suppose the r.v. X takes values in the set $\{x_1, x_2, \dots, x_n\}$
- Let p_i be the probability of $\mathrm{X} = x_i$.

$$0 \le p_i \le 1, \sum_{i=1}^{n} p_i = 1$$

- Expectation of X (the weighted average of X)

$$E[X] = p_1 x_1 + p_2 x_2 + \cdots + p_n x_n$$

- Variance of X (average squared variability)

$$Var[X] = p_1(x_1 - E[X])^2 + p_2(x_2 - E[X])^2 + \cdots + p_n(x_n - E[X])^2$$
$$= p_1 x_1^2 + p_2 x_2^2 + \cdots + p_n x_n^2 - (E[X])^2$$

- Standard deviation of X

$$Std[X] = \sqrt{Var(X)}$$

Probability, Expectation and Variance

- Multiplication of a random variable by a constant

$$E[cX] = cE[X]$$

$$Var(cX) = E[(cX - E[cX])^2] = c^2 Var(X)$$

- For two independent variables

$$Var(c_1 X_1 + c_2 X_2) = c_1^2 Var(X_1) + c_2^2 Var(X_2)$$

- Multiplication of two independent random variables

$$E[XY] = E[X]E[Y]$$

$$Var(XY) = E[(XY - E[XY])^2] = E[X^2Y^2 - 2XYE[XY] - E[XY]^2]$$

$$= E[X^2]E[Y^2] - (E[X]E[Y])^2$$

Expected PW

$$PW = C_0 + \frac{C_1}{1 + MARR} + \frac{C_2}{(1 + MARR)^2} + \cdots + \frac{C_N}{(1 + MARR)^N}$$

- In reality, C_0, C_1, C_2,…, C_N are random variables. Hence PW is also a random variable.

$$E[PW] = E[C_0] + \frac{E[C_1]}{1 + MARR} + \frac{E[C_2]}{(1 + MARR)^2} + \cdots + \frac{E[C_N]}{(1 + MARR)^N}$$

  - Step 1: Calculate each $E[C_i]$
  - Step 2: Discount the expected cash flows back using MARR

Example 8.7

- A company is considering the plan to launch a new product.
  - Initial cost $200,000.
  - Five year study period. No salvage value.
  - MARR=10%.

Price per unit	80
Variable cost per unit	60
Fixed cost	60,000

  - The units we can sell, denoted by Q, is random. Every year the probability distribution of Q is given by Pr(Q = 4,500) = 0.6 and Pr(Q = 7,300) = 0.4.
  - Calculate E[PW].

Example 8.7

- Solutions
  - For i=1,2,…,n
  - $C_i = (80 - 60)Q - 60000 = 20Q - 60000$
  - $E[C_i] = E[20Q - 60000] = 20E[Q] - 60000$
    $= 20(0.6 \times 4500 + 0.4 \times 7300) - 60000 = 52400$
  - $E[PW] = -200000 + 52400(P/A, 10\%, 5) = -1362.77$

How to make decisions using E[PW]

- Single project evaluation:
  - accept if E[PW] ≥ 0.
- Choose among multiple alternatives:
  - the best one has the highest E[PW].

Example 8.8

- 8-year project, MARR=12%
- If a failure occurs in a particular year
  - The cost is 80,000 with probability of 0.65
  - The cost is 120,000 with probability of 0.35

Failures are independent over years

Alternative	Investment	Annual maintenance cost	Prob. of failure in a single year
A	90,000	9,000	0.40
B	100,000	10,000	0.10
C	160,000	16,000	0.01

Example 8.8

- Alternative A, annual cost is
  - No failure: AW_0=90,000(A/P,12%,8)+9,000=27,117
  - Failure case 1: AW_1=AW_0+80,000=107,117
  - Failure case 2: AW_2=AW_0+120,000=147,117

Alternative	Investment	Annual maintenance cost	Prob. of failure in a single year
A	90,000	9,000	0.40
B	100,000	10,000	0.10
C	160,000	16,000	0.01

Example 8.8

- Expected annual cost for Alternative A

  $AW_0*0.6+AW_1*0.4*0.65+AW_2*0.4*0.35=64,717$

- Variance of annual cost for Alternative A

  $(AW_0-64,717)^2*0.6+ (AW_1-64,717)^2*0.26+$
  $(AW_2-64,717)^2*0.14=47,605^2$

- Question: which one to choose?

Alternative	Investment	Annual maintenance cost	Prob. of failure	Expected annual cost	Variance
A	90,000	9,000	0.40	64,717	$47,605^2$
B	100,000	10,000	0.10	39,530	$28,838^2$
C	160,000	16,000	0.01	49,148	$9,545^2$

Example 8.9: Useful life may be random

- A project with MARR=12%
  - Initial investment=$521,000
  - Annual profit=$79,600
  - Useful life (*N*) is a discrete random variable
  - For each *N*, PW=−521,000+79,600(P/A,12%,*N*)

Useful life	12	13	14	15	16	17	18
Probability	0.05	0.1	0.2	0.3	0.2	0.1	0.05
PW	-27926	-9689	6605	21148	34130	45720	56076

  - E[PW]= 0.05(-27926)+0.1(-9689)+…+0.05(56076)=$19,500

Example 8.10: Interest rate may be random

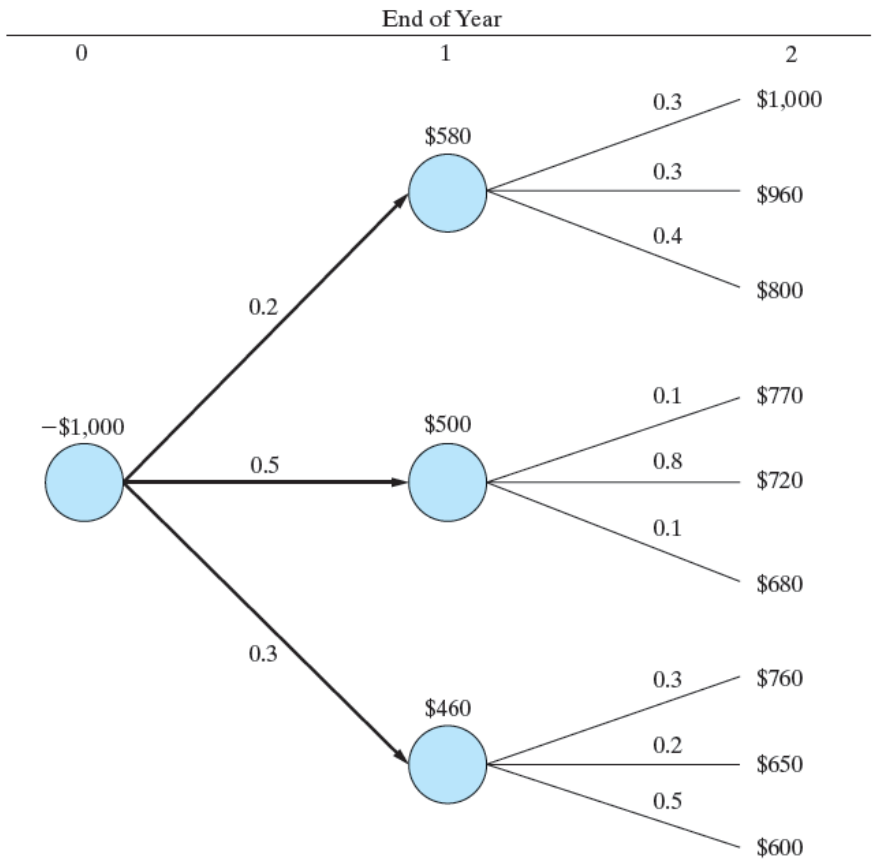
- $1M loan is to be repaid by a lump sum payment after three years
  - Annual interest rate in year 1: 3% fixed
  - Annual interest rate may be adjusted after one year, to 3%, 4%, or 5%, with equal probability
  - What is the expected lump sum payment?

Example 8.10: Interest rate may be random

- Method 1:
  - With 1/3 probability FW=1*1.03*1.03*1.03=1.092727M
  - With 1/3 probability FW=1*1.03*1.04*1.04=1.114048M
  - With 1/3 probability FW=1*1.03*1.05*1.05=1.135575M
  - E[FW] = 1.114117M
- Method 2:
  - Expected annual interest rate in years 2 and 3 is

    (3%+4%+5%)/3=4%
  - E[FW] = 1*1.03*1.04*1.04=1.114048M
- Which one is correct?

Random variables are not independent



Probability tree

The profit of year 2 depends on the profit of Year 1

Random variables are not independent

- There are 9 possibilities (MARR=12%)

Year 0	Year 1	Probability	Year 2	Probability	PW	Probability	PW^2
-1000	580	0.2	1000	0.3	315.051	0.06	99257.15
-1000	580	0.2	960	0.3	283.1633	0.06	80181.43
-1000	580	0.2	800	0.4	155.6122	0.08	24215.17
-1000	500	0.5	770	0.1	60.26786	0.05	3632.215
-1000	500	0.5	720	0.8	20.40816	0.4	416.4931
-1000	500	0.5	680	0.1	-11.4796	0.05	131.781
-1000	460	0.3	760	0.3	16.58163	0.09	274.9505
-1000	460	0.3	650	0.2	-71.1097	0.06	5056.589
-1000	460	0.3	600	0.5	-110.969	0.15	12314.21
					Expected PW=	39.52487	
					Expected (PW^2)=	15233.6	
					Var=	13671.38	

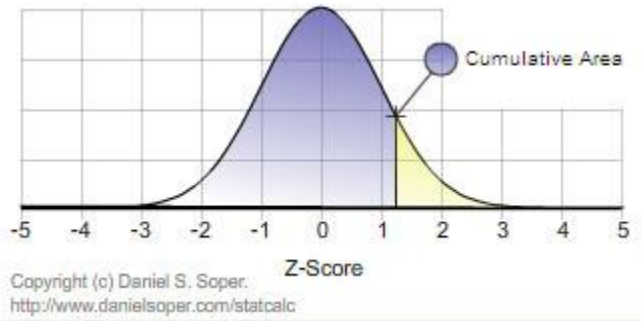
Random variables are not independent

- More information
  - Variance: $Var(PW) = E[PW^2] - (E[PW])^2 = 13,671$
  - Standard deviation: $Std(PW) = (13671)^{0.5} = 117$
- What is the probability that the project cannot earn money?
  - $Prob(PW<0) = 0.05+0.06+0.15 = 0.26$
- Which analysis result(s) favor approval of the project, and which ones appear unfavorable?
  - Mean
  - Probability
  - Variance/Standard deviation

Normal distribution $X \sim N(\mu, \sigma^2)$

- Mean = μ
- Variance = σ^2
  - Standard deviation = σ
- Standard normal distribution Z ~ N(0, 1)
  - If $X \sim N(\mu, \sigma^2)$, then $(X-\mu)/\sigma \sim N(0,1)$, i.e., $X = \mu + Z\sigma$
- Probability of an event

  Prob $(X<c)$ = Prob $(\mu+Z\sigma<c)$ = Prob$(Z<(c-\mu)/\sigma)$



Normal distribution $X \sim N(\mu, \sigma^2)$

- If $X \sim N(10, 25)$, then $X = 10 + 5Z$
- Probability of an event

Prob $(X<13.6)$ = Prob $(Z<(13.6-10)/5)$

= Prob($Z<0.72$) = 0.7642

Prof $(X>8)$ = 1 − Prob $(X<8)$ = 1 − Prob $(Z<(8-10)/5)$

= 1 − Prob $(Z<-0.4)$ = 1 − 0.3446 = 0.6554

TABLE E-1 Areas under the Normal Curve (cont'd.)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133

Properties of normal distribution

- If $X_1 \sim N(\mu_1, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$, and X_1 and X_2 are independent, then
  - The sum: $X_1+X_2 \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$
  - The difference: $X_1-X_2 \sim N(\mu_1-\mu_2, \sigma_1^2+\sigma_2^2)$
- If $X_1,X_2,\ldots,X_n$ are independent and n is large, then the sum $k_1X_1+k_2X_2+\ldots+k_nX_n$ approximates to normal distribution, $N(\mu, \sigma^2)$, with
  - Mean $\mu = k_1\mathrm{E}[X_1]+k_2\mathrm{E}[X_2]+\ldots+k_n\mathrm{E}[X_n]$
  - Variance $\sigma^2 = k_1^2\mathrm{Var}[X_1]+k_2^2\mathrm{Var}[X_2]+\ldots+k_n^2\mathrm{Var}[X_n]$

Present worth with normal distribution

- The cash flows for each year F_0, F_1, …, F_n, are normally distributed independent variables
  - $PW=F_0+F_1(P/F,i\%,1)+\ldots+F_n(P/F,i\%,n)$
  - $E[PW]=E[F_0]+E[F_1](P/F,i\%,1)+\ldots+E[F_n](P/F,i\%,n)$
  - $Var[PW]=Var[F_0]+Var[F_1](P/F,i\%,1)^2+\ldots+Var[F_n](P/F,i\%,n)^2$

Example 8.11

- Normal distribution

Year k	mean	Standard dev
0	-7000	0
1	3500	600
2	3000	500
3	2800	400

- E[PW]= -7000+3500(P/F,15%,1) +…+2800(P/F,15%,3) =153
- Var[PW]= $0+600^2(P/F,15\%,1)^2+\ldots+400^2(P/F,15\%,3)^2=484{,}324$
- PW is normally distributed $N(153,696^2)$
  - the sum of independent normal random variables
- The probability that PW<0 is

$$\text{Prob}(PW<0) = \text{Prob}(Z\sigma+\mu<0) = \text{Prob}(Z<-\mu/\sigma)$$
$$= \text{Prob}(Z<-0.22)=0.4129$$

Normal approximation

- $PW = F_0 + F_1(P/F, i\%, 1) + \ldots + F_n(P/F, i\%, n)$
  - When F_k are independent, n is large
  - PW can be regarded as a normal distribution, even if we do not know the distribution of F_i
  - $E[PW] = E[F_0] + E[F_1](P/F, i\%, 1) + \ldots + E[F_n](P/F, i\%, n)$
  - $Var[PW] = Var[F_0] + Var[F_1](P/F, i\%, 1)^2 + \ldots + Var[F_n](P/F, i\%, n)^2$

Example 8.12

- Cash flow F_k depends on two r.v. X_k and Y_k

		Mean		Standard dev		Mean of	Var of
Year	F_k	X_k	Y_k	X_k	Y_k	F_k	F_k
0	X_0+Y_0	0	−100000	0	10000	−100000	$100*10^6$
1	X_1+Y_1	60000	−20000	4500	2000	40000	$24.25*10^6$
2	X_2+2Y_2	65000	−15000	8000	1200	35000	$69.76*10^6$
3	$2X_3+3Y_3$	40000	−9000	3000	1000	53000	$45.0*10^6$
4	X_4+2Y_4	70000	−20000	4000	2000	30000	$32.0*10^6$
5	$2X_5+2Y_5$	55000	−18000	4000	2300	74000	$85.16*10^6$

Example 8.12

- E[PW]= −100000+40000(P/F,20%,1)+… =$32517
- Var[PW]= $100*10^6+24.25*10^6(P/F,20\%,1)^2+\ldots=186.75*10^6$
- So, PW~$N(32517,13666^2)$
- Prob(PW<0)=Prob(Z<−32517/13666)=Prob(Z<−2.3794)

		Mean		Standard dev		Mean of	Var of
Year	F_k	X_k	Y_k	X_k	Y_k	F_k	F_k
0	X_0+Y_0	0	−100000	0	10000	−100000	$100*10^6$
1	X_1+Y_1	60000	−20000	4500	2000	40000	$24.25*10^6$
2	X_2+2Y_2	65000	−15000	8000	1200	35000	$69.76*10^6$
3	$2X_3+3Y_3$	40000	−9000	3000	1000	53000	$45.0*10^6$
4	X_4+2Y_4	70000	−20000	4000	2000	30000	$32.0*10^6$
5	$2X_5+2Y_5$	55000	−18000	4000	2300	74000	$85.16*10^6$

Sequential decisions over time

- Previously in handling uncertainties
  - We assume a **single** decision is to be made at the beginning
- Sometimes, we need to make several decisions sequentially over time
  - The outcome of each decision may be random
  - The goal is to optimize the expected value for the overall decisions
- Decision sequences
  - Decision 1→outcome 1 (random)
  - Knowing outcome 1→decision 2→outcome 2 (random)
  - Outcome 2→decision 3→outcome 3 (random)
  - ……
- Decision tree will be a useful tool

Example 8.13: Decision tree for single-step decision

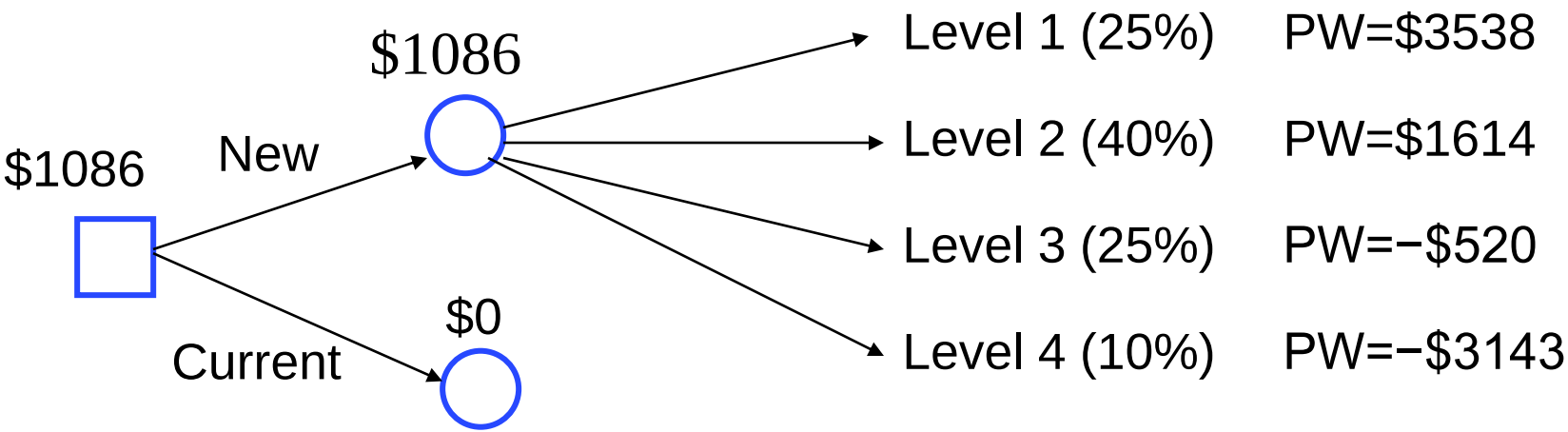
- A new design is proposed to replace the current one
  - Decision: whether to take the new design
  - Four possible outcomes for the new design
- MARR=18%, study period=6 years
- Initial invest=8600

Service level	Probability	Annual saving	PW
1	0.25	3,470	3,538
2	0.40	2,920	1,614
3	0.25	2,310	-520
4	0.10	1,560	-3,143

Expected PW for taking the new design is $1,086.

Example 8.13: Decision tree for single-step decision

- Two types of nodes
  - Decision nodes □ and Outcome nodes ○
- Identify all possible outcomes for each decision
  - With probabilities
- Calculate E[PW] for each node in a backward way
  - Outcome node: using probability to obtain the expectation
  - Decision node: making comparison to obtain the best

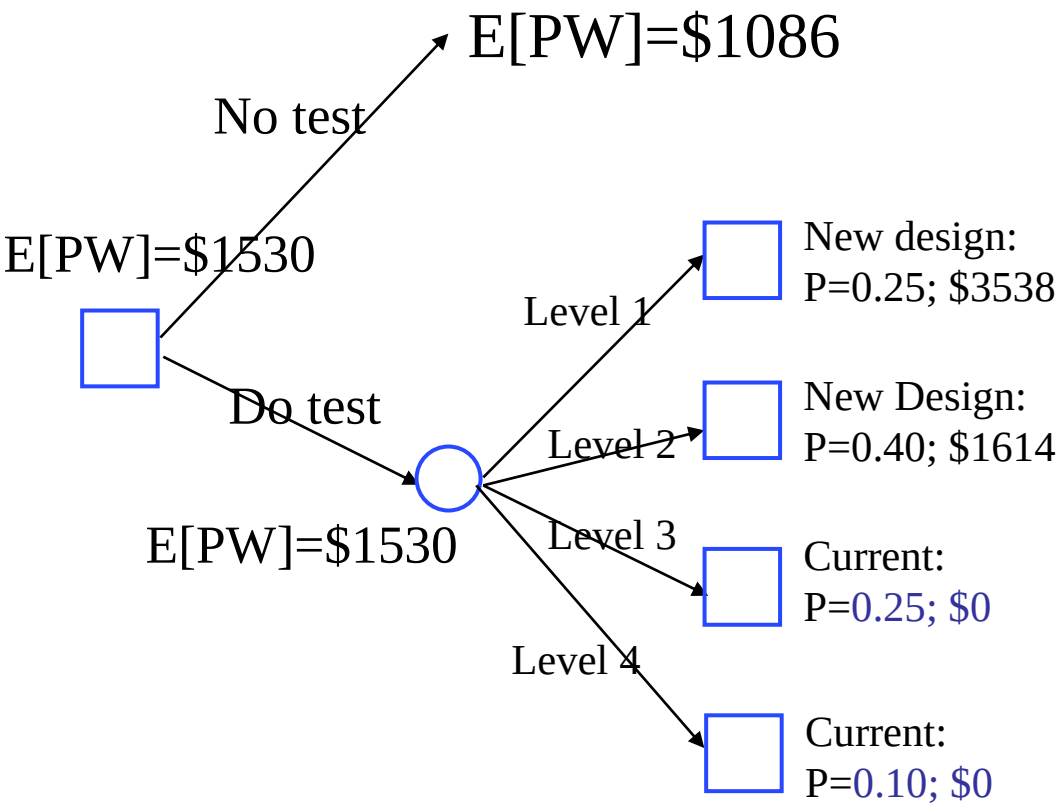


Example 8.14: Two-step decision making

- If we know what is going to happen to the new design, we can have a better decision
  - How?
- A test can be conducted on the new design
  - Two-step decision
    - (1) whether to make the test first
    - (2) whether to take the new design (after knowing the test result)
- Suppose that the test can predict the future accurately
  - Perfect information

Example 8.14: Two-step decision making

- A test can tell the service levels with certainty



Conclusion:

Do the test first

Expected Value of Perfect Information

EVPI=1530−1086=$444

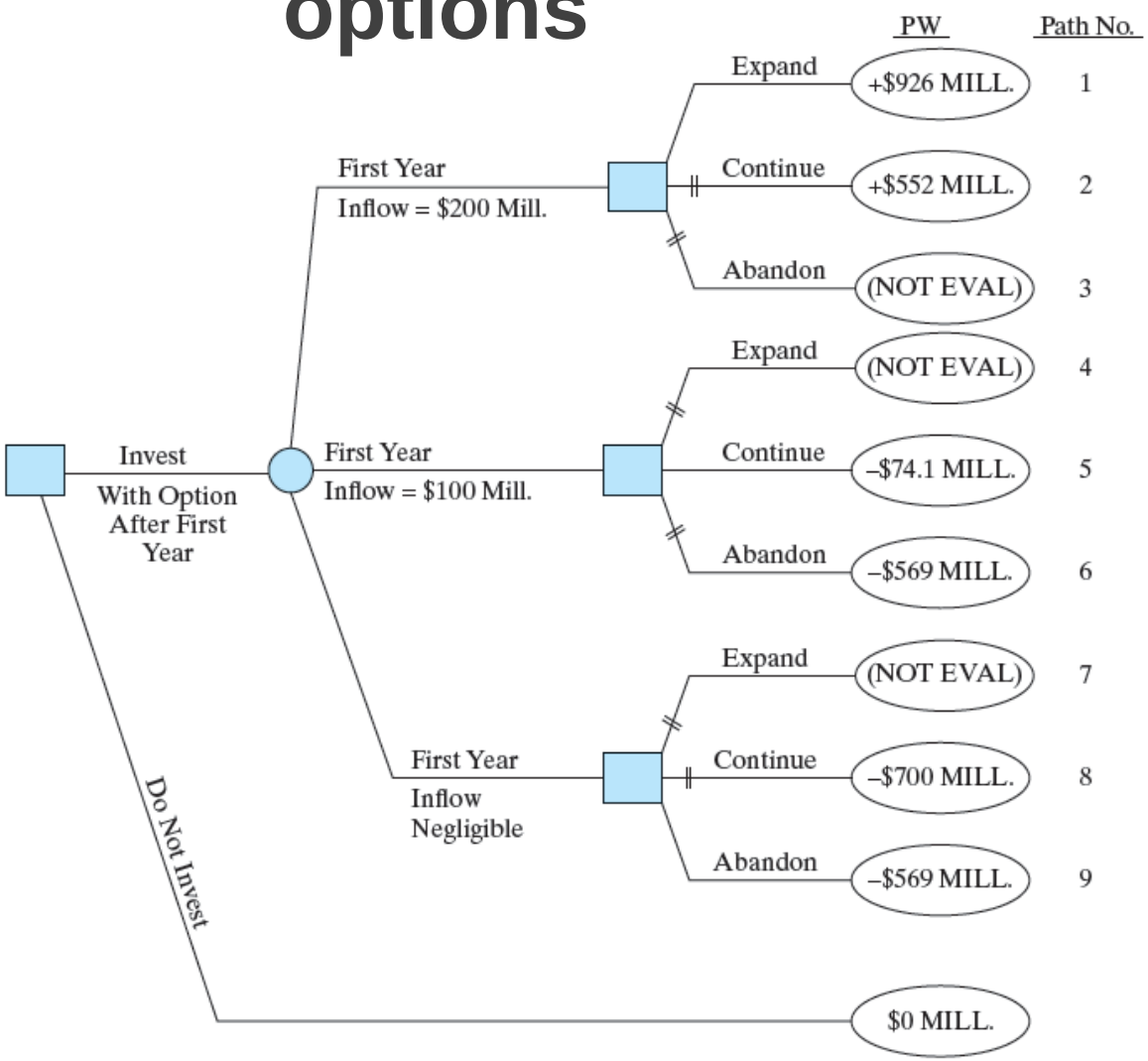
Example 8.15: Decision tree with real options

- A project has the information below
  - Initial investment = $700M, Useful life = 20 years, MARR = 15%
  - Random annual income with identical probability
    - $200M (high demand, capacity not enough)
    - $100M (normal demand, capacity enough)
    - $0M (no demand, capacity not used)
- By one-step decision

  Average annual income = (200+100+0)/3 = $100M

  PW=-700+100(P/A,15%,20) = -74.1M
- Options after one year, after observing the first-year output
  - To expand: investment = $500M, extra annual income = $150
  - To continue: do nothing
  - To abandon: salvage value = $150
  - How may the options help?

Example 8.15: Decision tree with real options



Example 8.15: Decision tree with real options

First-year	Option	Equation	PW
high demand	Expand	-700+200(P/A,15%,20)-500(P/F,15%1)+150(P/A,15%,19)(P/F,15%,1)	926
	Continue	-700+200(P/A,15%,20)	552
	Abandon		
Normal demand	Expand		
	Continue	-700+100(P/A,15%,20)	-74
	Abandon	-700+(100+150)(P/F,15%,1)	-483
No demand	Expand		
	Continue	-700	-700
	Abandon	-700+150(P/F,15%,1)	-569

Example 8.16: Risk-adjusted interest rate

- Banks charge a high interest rate to risky borrowers.
- A small company borrows $1 million from a bank
  - (uniform) annual repayment for 5 years.
- The bank classifies this company to be risky, where in any year there is 5% probability that the company may be bankrupted.
  - If the company is bankrupted within the first year, the bank can receive zero repayment
  - If the company is bankrupted within the second year, the bank can receive the first repayment only
- Assume that the bank charges 3% interest rate to reliable borrowers. What is the lowest interest rate that the bank should charge to this risky company?

Example 8.16

- Let A be the annual repayment
- 5% probability, the company bankrupted within the 1st year:
  - Cash flow: (-1M, 0, 0, 0, 0, 0),
  - $PW_1 = -1$
- (1-5%)*5% probability, the company bankrupted within the 2nd year:
  - Cash flow: (-1M, A, 0, 0, 0, 0),
  - $PW_2 = -1 + A(P/A, 3\%, 1) = -1 + 0.9709A$
- $(1-5\%)^2$ *5% probability, the company bankrupted within the 3rd year:
  - Cash flow: (-1M, A, A, 0, 0, 0)
  - $PW_3 = -1 + A(P/A, 3\%, 2) = -1 + 1.9135A$

Example 8.16

- $(1\text{-}5\%)^3$ *5% probability, the company bankrupted within the 4rd year:
  - Cash flow: (-1M, A, A, A, 0, 0)
  - PW_3 = -1 + A(P/A, 3%, 3) = -1 + 2.83A
- $(1\text{-}5\%)^4$ *5% probability, the company bankrupted within the 5th year:
  - Cash flow: (-1M, A, A, A, A, 0),
  - PW_5 = -1 + A(P/A, 3%, 4) =-1 + 3.7171A
- $(1\text{-}5\%)^5$ probability, the company not bankrupted within the 6th year:
  - Cash flow: (-1M, A, A, A, A, A)
  - PW_6 = -1 + A(P/A, 3%, 5) =-1 + 4.5797A

Example 8.16

- $E[PW] = PW_1*5\% + PW_2*0.95*0.05$
  $+ PW_3*0.95^2*0.05 + \ldots + PW_6*0.95^5$
  $= -1 + 3.9488A$
- To have a non-negative E[PW], we need
  - $A \geq 0.2532$ → i=rate(5, 0.253241, -1) = 8.42%

Example 8.17

- A new product that has 5% probability of failure.
  - If the product is successful, its IRR will be x%
  - If the product fails, all investment becomes wasted.
- What is the minimum value of x acceptable to the company if the company has an MARR of 3%?

Example 8.17

- If it is a one-year project, we should have

  $0.95(1+x)=1+3\% \rightarrow x=8.42\%$

- If it is a two-year project (with no return in year 1),

  $0.95(1+x)^2 = (1+3\%)^2 \rightarrow x=5.68\%$

- If it is a two-year project with each annual return A (no salvage value),

  $0.95(A/1.03+A/1.03^2) = 1 \rightarrow$

  $A=0.55 \rightarrow x=\text{rate}(2, 0.55, -1) = 6.61\%$

Dealing with Multi-attributes

Multi-Attributes

- A project or decision may have various consequences
  - Not all can be evaluated by dollars
- Some attributes can be used to trade off with each other
  - Compensatory models
- Some attributes cannot be traded off with each other
  - Non-compensatory models

Example 8.18: equipment selection

	A	B	C	Current
Capital invest	$115000	$338950	$32000	$0
Reduction of design time	60%	67%	50%	0
Flexibility	Excellent	Excellent	Good	Poor
Quality	Excellent	Excellent	Good	Poor
…				

Non-compensatory models

- Different attributes cannot be compared directly
  - Dominance
  - feasible ranges
  - lexicography

Dominance

- Dominance
  - Alternative A dominates Alternative B if
    - (1) A is no-worse than B with respect to every attribute, and
    - (2) A is better than B with respect to at least one attribute
  - Alternative B, if dominated by A, should be discarded
- Example:
  - Alternative A: E[PW]=\$1000, Var[PW]=300^2
  - Alternative B: E[PW]=\$900, Var[PW]=300^2
  - Alternative C: E[PW]=\$900, Var[PW]=100^2
  - A dominates B, C dominates B

Using feasible ranges

- Feasible ranges
  - Each attribute is assigned a minimum or maximum acceptable value
- Satisficing
  - An alternative is discarded if **at least** one attribute cannot achieve the acceptable value
- Disjunctive resolution
  - An alternative is discarded if **no** attribute can achieve the acceptable value

Using feasible ranges

- Example:
  - Alternative A: E[PW]=\$1000, Var[PW]=300^2
  - Alternative B: E[PW]=\$900, Var[PW]=300^2
  - Alternative C: E[PW]=\$900, Var[PW]=100^2

  - If we decide E[PW]>950, only A is left by satisficing
  - If we decide E[PW]>950, Var[PW]<200^2, B is discarded by disjunction resolution

Lexicography method

- Assuming that attributes can be ranked from the most important to the least important
- We first compare alternatives with respect to the most important attribute, and select the best
- If such best alternatives are not unique, we look at the second important attribute, and choose the best from them
- ……

Example 8.19: Selecting a dentist

Dr.	Molar	Feelgood	Whoops	Pepper
Cost ($/hr)	50	80	20	40
Technique	A	C	D	B
Distance	15	20	5	30
Office hours	40	25	40	40
Work quality	Excellent	Fair	Poor	Good

- Dominance
  - Dr. Molar dominates Dr. Feelgood
  - Dr. Feelgood is eliminated
- Dominance should be checked for each pair

Example 8.19: Selecting a dentist

					Acceptable value	
Dr.	Molar	Feelgood	Whoops	Pepper	min	Max
Cost ($/hr)	50	80	20	40		60
Technique	A	C	D	B	C	
Distance	15	20	5	30		30
Office hours	40	25	40	40	30	
Work quality	Excellent	Fair	Poor	Good	Good	

- By satisficing
  - Dr. Whoops is eliminated for the attribute of Technique level/work quality
  - Dr. Feelgood is eliminated for the attribute of office hours
  - Drs. Molar and Peppers are left
- By disjunctive resolution
  - No alternatives are eliminated in this example

Example 8.19: Selecting a dentist

Dr.	Molar	Feelgood	Whoops	Pepper
Cost ($/hr)	50	80	20	40
Technique	A	C	D	B
Distance	15	20	5	30
Office hours	40	25	40	40
Work quality	Excellent	Fair	Poor	Good

- Suppose the attributes are sorted from the most important as follows
  - Work quality>cost>technique>Office hours>distance
- Looking at the attribute “Work quality”
  - Dr. Molar should be selected
- If the work quality of Dr. Pepper is also “excellent”
  - Both Drs. Molar and Pepper have the same best value with respect to the most important attribute
  - Comparing cost, Dr. Pepper should be selected

Compensatory models

- Basic assumption
  - Good values of one attribute can be used to compensate poor values of other attributes
  - A unified scaling for each attribute is needed
- Non-dimensional scaling
  - To convert the value of each attribute to an index in [0,1]
  - The most desirable value corresponds to “1”
  - The least desirable value corresponds to “0”
  - Other values are linearly assigned within [0,1]

Non-dimensional scaling

- Cost
  - Min value: $20; Max value: $80
  - For any cost: (80−cost)/(80−20)=(80−cost)/60
- Technique
  - Best value: A→1; worst value: D→0; B→2/3; C→1/3
- Distance
  - Min value: 5; max value: 30; (30−distance)/25
- Office hours
  - Min value: 25; max value: 40; (office hours−25)/15
- Quality
  - Excellent: 1; Good: 2/3; Fair: 1/3; Poor: 0

Example 8.20: Non-dimensional scaling

Dr.	Molar	Feelgood	Whoops	Pepper
Cost ($/hr)	0.5	0	1	0.67
Technique	1	0.33	0	0.67
Distance	0.6	0.4	1	0
Office hours	1	0	1	1
Work quality	1	0.33	0	0.67
sum	4.1	1.06	3	3

- A simple way to compare
  - Using the sum of all scaled index
  - Dr. Molar is the best

Additive weighting technique

- Each attribute *j* is assigned a weight w_j
  - $w_1+w_2+\ldots+w_m=1$, $w_j>0$
- We use the weighted sum of indices for all attributes to make decisions
  - The problem is converted to a single-attribute problem
- How the weights are determined?
  - By experience
  - By systematic method

Example 8.20: Determining weights

- A systematic method
  - Ranking the attributes according to their relative importance
  - 1: the least importance
  - The weight of each attribute *j* is given by

    Importance index *j* / sum of importance indices

Attributes	Importance	Weights	Molar	Feelgood	Whoops	Pepper
Cost ($/hr)	4	4/15	0.5	0	1	0.67
Technique	3	3/15	1	0.33	0	0.67
Distance	1	1/15	0.6	0.4	1	0
Office hours	2	2/15	1	0	1	1
Work quality	5	5/15	1	0.33	0	0.67
	Sum=15		0.84	0.21	0.47	0.66

- Chapters 11, 12, 13 of the course textbook.